PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DAR ES SALAAM REGION
FORM SIX MOCK EXAMINATION
CHEMISTRY 3A

Code: 132/3A

Time: 3:00 Hours — Friday 16th February, 2024 A.M.

INSTRUCTIONS

1. This paper consists **three (3)** question.
2. Answer **all** questions.
3. Question number 1 carries **20 marks** and other two (2), **15 marks** each.
4. All answers must be written in the answer booklets provided
5. Mathematical tables and non – programmable calculators may be used.
6. Cellular phones and other unauthorized materials are not allowed in the examination room
7. Write you **Examination Number** on every page of your answer booklet (s)

Atomic masses: H = 1, C = 12, N = 14, O = 16, Na = 23, Cl = 35.5, Mn = 55, K = 39, S = 32

Molar gas constant R = $8.314 Jk^{-1} mol^{-1}$ or $0.821 Jmol^{-1} k^{-1}$

1. You are provided with the following

N : A solution made by diluting 12.5cm³ of a given hydrogen peroxide with 500cm³ of distilled water

P : A standard solution of 0.02M potassium permanganate

Q : A 2M sulphuric acid.

Theory

The concentrations of commercial hydrogen peroxide are usually expressed as volume strength. Volume strength refers to the volume of oxygen in litres at STP which would be liberated when one mole of hydrogen peroxide decomposes as per equation below.

$$2H_2O_{2(aq)} \rightarrow 2H_2O_{(l)} + O_{2(g)}$$

Hydrogen peroxide reacts with acidified potassium permanganate according to the following equation.

$$MnO^-_{4(aq)} + H_2O_{2(aq)} + H^+_{(aq)} \rightarrow Mn^{2+}_{(g)} + O_{2(g)}$$

Procedures.

(i) Pipette 20cm³ (or 25 cm³) of solution N into a conical flask. Add to it about 20cm³ of Q.

(ii) Titrate this mixture with solution P until permanent pink colour is observed.

(iii) Repeat this procedure (i) to (ii) obtain three more reading. Tabulate your results as shown in the table below

The volume of the pipette uses was ________ cm³

Titration number	Pilot	1	2	3
Final reading (cm³)				
Initial readings (cm³)				
Titre value (cm³)				

Summary.

_________cm³ of solution N required______cm³ of solution P for complete reaction

Questions.

(a) Write half reaction equations and overall equation to show oxidations and reduction process.

(b) Explain why at the end point the solution in a conical flask was pink in colour.

(c) Calculate the:-

(i) Concentration of hydrogen peroxide in mol/dm³

(ii) Concentration of hydrogen peroxide in g/dm³

(iii) Original concertation of hydrogen peroxide in g/dm³

(iv) Volume strength of commercial hydrogen peroxide.

2. You are provided with following solution

H_1 : A solution of 0.5M $Na_2S_2O_3$

H_2 : A solution of 0.1M Hcl

Stop watch / stop clock

Thermometer (0°C – 100°C)

Theory

A white precipitate of amorphous Sulphur can be obtained by the action of dilute acid on sodium thiosuphate according to the equation

$$S_2O^{2-}_{3(aq)} + 2H_3O^+_{(aq)} \rightarrow 3H_2O_{(l)} + S_{(S)} + SO_{2(g)}$$

The precipitated Sulphur causes the solution to become opaque. From this phenomenon, you can assess the rate of Sulphur precipitated by measuring the time taken for the solution to become totally opaque by Sulphur.

Procedures.

(i) Use a blue/ black pointed pen draw a letter X on a piece of white paper and place a small beaker on top of letter X such that the letter X is visible through the solution

(ii) Prepare water bath

(iii) Measure 10cm³ of H_1, and 10cm³ of H_2 into separate boiling tubes.

(iv) Put the two boiling tubes containing solution H_1 and H_2 in to water bath and warm the content to about 50°C.

(v) Immediately pour the hot solution H_1 and H_2 into a small beaker on top of letter X and simultaneously start the stop watch. Record the time of disappearance of letter X through your eyes.

(vi) Repeat the whole procedure using the temperature 60°C, 70°C and 80°C Record your result in a table form as shown below.

Temperature		time	$^1/_t$	Log $^1/_t$	$^1/_T$
(°C)	K	(sec)	(sec⁻¹)	(sec⁻¹)	(K⁻¹)
50					
60					
70					
80					

Question

(a) Plot a graph of log $^1/_t$ against $^1/_T$ (K⁻¹)

(b) Determine the slope of the graph obtained in (a) above

(c) Determine the actvation energy (Ea)

(d) Determine the value of pre – exponential factor (A)

3 Sample M is simple salt containing one cation and one anion. Carefully carry out qualitative analysis experiment to identify the ions present in the salt based on the following tests.

(a) Appearance of the sample

(b) Actions of heat on the solid sample

(c) Solubility

(d) Action of aqueous NaOH on a solution M

(e) Action of fresh prepare $FeSO_4$ solution on solution of M followed by conc. H_2SO_4 through the side of the test tube.

(f) Action of lead ethanoate and then boil

(d) Perfom a confirmation test for the cation and anion.

Questions

(i) Prepare relevant table showing the qualitative analysis results.

(ii) Write a balanced chemical equation for the reaction in experiment(b)

(iii) The cation present in sample M was _______and the anion was________

(iv) Write chemical formula of the sample M.

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CHEMISTRY 3A

MARKING SCHEME

1. The Volume of the pipette Used was 25 cm^3 $0\frac{1}{2}$

Table of Results:

Titration number	Pilot	1	2	3
Final reading (cm^3)	15.60	15.50	15.40	15.40
Initial reading (cm^3)	0.00	0.00	0.00	0.00
Titre value (cm^3)	15.60	15.50	15.40	15.40

$04\frac{1}{2}$

T = $01\frac{1}{2}$
A.C = 02
D.P = 01

$$\text{Average Volume used} = \left(\frac{V_1 + V_2 + V_3}{3}\right) cm^3$$

$$= \left(\frac{15.50 + 15.40 + 15.40}{3}\right) cm^3$$

$$= 15.43\ cm^3$$

∴ The average Volume used = 15.43 cm^3 01

Summary:

25.00 cm^3 of Solution N required 15.43 cm^3 of Solution P for Complete reaction. 01

(a) Reduction half reaction:

$$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O \quad 0\tfrac{1}{2}$$

Oxidation half reaction:

$$H_2O_2 \longrightarrow O_2 + 2H^+ + 2e^- \quad 0\tfrac{1}{2}$$

Overall reaction equation:

$$2MnO_4^- + 5H_2O_2 + 6H^+ \longrightarrow 2Mn^{2+} + 5O_2 + 8H_2O \quad 01$$

(b) (i) Concentration of H_2O_2 In mol/dm^3

- From the equation:

$$2MnO_4^- + 5H_2O_2 + 6H^+ \longrightarrow 2Mn^{2+} + 5O_2 + 8H_2O$$

$M_{MnO_4^-} = 0.02M$

$V_{MnO_4^-} = 15.43cm^3$

$V_{H_2O_2} = 25cm^3$

$n_{MnO_4^-} = 2$

$n_{H_2O_2} = 5$

$M_{H_2O_2} = ?$

Then;

$$\frac{M_{MnO_4^-} \cdot V_{MnO_4^-}}{M_{H_2O_2} \cdot V_{H_2O_2}} = \frac{n_{MnO_4^-}}{n_{H_2O_2}}$$ 01/2

$$M_{H_2O_2} = \frac{M_{MnO_4^-} \cdot V_{MnO_4^-} \cdot n_{H_2O_2}}{V_{H_2O_2} \cdot n_{MnO_4^-}}$$

$$= \frac{0.02M \times 15.43\,cm^3 \times 5}{25cm^3 \times 2}$$ 01/2

$$= 0.03086M$$

∴ The Concentration of H_2O_2 = 0.03086M 01

(ii) Concentration of H_2O_2 in g/dm^3

From:

$$\text{Molarity} = \frac{\text{Concentration } (g/dm^3)}{\text{Molar mass}} \quad 0\tfrac{1}{2}$$

$$\text{Concentration } (g/dm^3) = \text{Molarity} \times \text{Molar mass}$$

$$= 0.03086\, mol/dm^3 \times 34\, g/mol \quad 0\tfrac{1}{2}$$

$$= 1.049\, g/dm^3$$

$\therefore$ The Concentration of $H_2O_2 = 1.049\, g/dm^3$ 01

(iii) Original Concentration of H_2O_2 in g/dm^3

$M_d = 0.03086\,M$

$V_d = 512\, cm^3$

$M_c = ?$

$V_c = 12.5\, cm^3$

From Dilution law;

$$M_cV_c = M_dV_d \quad 0\tfrac{1}{2}$$

$$M_c = \frac{M_dV_d}{V_c}$$

$$= \frac{0.03086\,M \times 512\, cm^3}{12.5\, cm^3} \quad 0\tfrac{1}{2}$$

$$M_c = 1.264\,M \quad 01$$

But;

Concentration = Molarity × Molar Mass $(01/2)$

$= 1.264\,M \times 34\,g/mol$ $(01/2)$

$= 42.98\,g/mol$

∴ The Original Concentration of $H_2O_2 = 42.98\,g/mol$ (01)

(iv) Volume strength of H_2O_2

From the equation of decomposition of H_2O_2:

$$2H_2O_2 \longrightarrow 2H_2O + O_2 \quad (01/2)$$

From the equation above;

2 moles of $H_2O_2 \equiv$ 1 mol of O_2

but 1 mol of O_2 at STP $= 22.4\,dm^3$

and 2 mol of $H_2O_2 \equiv 2 \times 34g = 68g$ $(01/2)$

Then;

68g of $H_2O_2 \equiv 22.4\,dm^3$

42.98g of $H_2O_2 \equiv x\,dm^3$ (01)

$$x = 14.16\,dm^3$$

∴ The Volume strength of $H_2O_2 = 14.16\,dm^3$ (01)

2. Table of Results:

Temperature °C	Temperature K	Time (s)	$1/t$ (sec^{-1})	$\log \frac{1}{t}$ (s^{-1})	$1/T$ (K^{-1})
50	323	43	0.0233	−1.623	3.09×10^{-3}
60	333	20	0.0500	−1.301	3.00×10^{-3}
70	343	10	0.1000	−1.000	2.92×10^{-3}
80	353	5	0.2000	−0.69	2.83×10^{-3}

@ cell = 01 mark = 05 Marks.

(b) Slope of the graph:

$$\text{Slope} = \frac{\Delta \log 1/t\ (sec^{-1})}{\Delta \frac{1}{T}\ (K^{-1})} \qquad (0\tfrac{1}{2}\text{ mark})$$

$$= \frac{-1.46-(-0.46)\ sec^{-1}}{(3.10-2.77)\times10^{-3}\ K^{-1}} \qquad (0\tfrac{1}{2}\text{ mark})$$

$$= -3.21\times10^{-3}\ K(sec^{-1})$$

$\therefore$ The slope $= -3.21\times10^{-3}\ K(sec^{-1})$ (01 mark)

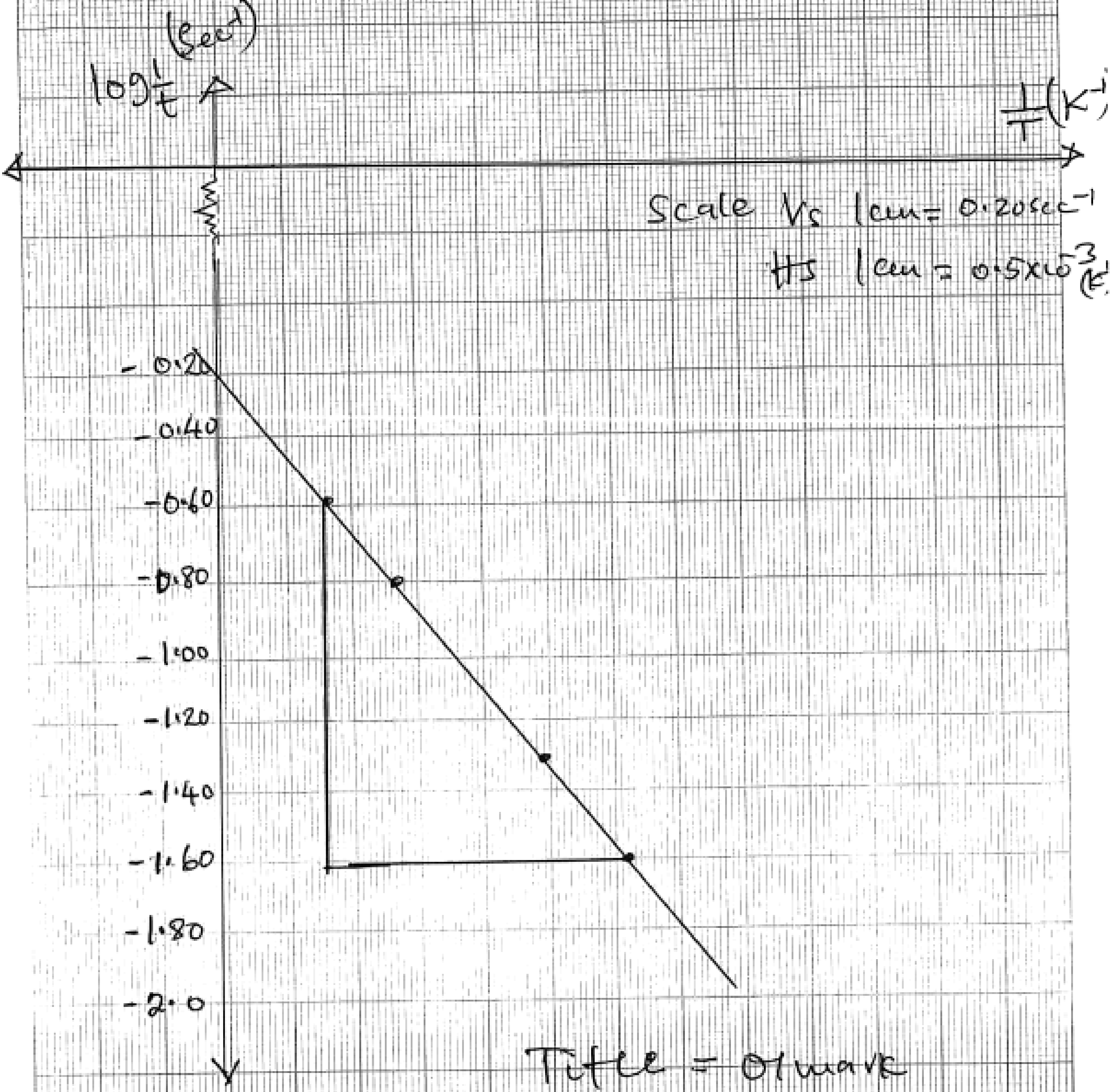
(C) Activation energy (Ea).

Arrhenius eqn;

$$\log\frac{1}{t} = \log A - \frac{Ea}{2.303R}\times\frac{1}{T} \qquad (01\text{ mark})$$

$$\text{Slope} = -\frac{Ea}{2.303R}$$

(a) GRAPH OF $\log\frac{1}{t}$ AGAINST $\frac{1}{T}$



Title = 01 mark
Scale = 01 mark
labelling of axis = 01 mark
best line = 01 mark

(1cm x 1cm) squares of (10mm x 10mm)

$$E_a = -\text{slope} \times 2.303\,R$$
$$= -(-3.21 \times 10^{-3} \times 2.303 \times 8.314) \text{ (01 mark)}$$
$$= 0.0615\ \text{KJ/mol}$$

$\therefore$ Activation energy $(E_a) = 0.0615\ \text{KJmol}^{-1}$ (01 mark)

(d) Arrhenius Constant (A).

From Arrhenius equation:

$$\underset{y}{\log \frac{1}{t}} = \underset{-m}{-\frac{E_a}{2.303R}} \times \underset{x}{\frac{1}{T}} + \underset{c}{\log A}$$

From the graph, $c = -0.2$.

Then; $A = \log^{-1}(-0.2)$

$$= 0.63\ \text{sec}^{-1}$$

$\therefore$ Arrhenius Constant $(A) = 0.63\ \text{sec}^{-1}$

(01 mark)

3.

S/N	Experiment	Observation	Inference
(a)	Appearance of the sample	White Crystalline salt (00½ mark)	- Non-transitional metals may be present. - NO_3^-, SO_4^{2-}, Cl^-, $C_2O_4^{2-}$, CrO_4^{2-}, NO_3^-, CH_3COO^-, $Cr_2O_7^{2-}$ may be present (00½ mark)
(b)	Action of heat on the solid sample	A white sublimate and a colourless gas with chocking smell which turns moist litmus paper from red to blue. (00½ mark)	NH_4^+ may be present (00½ mark)
		Colourless gas evolved which turn blue litmus paper to red and form dense white fumes with ammonia (00½ mark)	Cl^- may be present (00½ mark)
(c)	Solubility of Sample M	Sample was soluble in cold water (00½ mark)	Soluble salts may be present (00½ mark)
(d)	Action of aqueous NaOH on Solution M	No precipitate formed (00½ mark)	NH_4^+ may be present (00½ mark)
(e)	Action of fresh prepared $FeSO_4$ followed by Conc. H_2SO_4	No brown ring formed (00½ mark)	NO_3^- absent (00½ mark)
		A gas was formed which turns moist blue litmus paper red and form a dense white fumes on adding Conc. H_2SO_4	Cl^- may be present

S/N	Experiment	Observation	Inference.
(f)	Action of Lead ethanoate and boiled.	White ppt. formed which dissolve slowly on boiling ($00\frac{1}{2}$ mark)	Cl^- may be present. ($00\frac{1}{2}$ mark)
(g)	To a small solid sample, dil NaOH was added, then warmed. A glass rod dipped in a Conc. HCl passed to the mouth of the test tube.	Colourless gas which form white fumes with Conc. HCl and turn moist red litmus to blue evolved (01 mark)	NH_4^+ Confirmed (01 mark)
	To a small volume of sample solution M, dil. HNO_3 was added followed by addition of $AgNO_3$ then ammonia solution.	White precipitates soluble in dilute ammonia solution was observed. (01 mark)	Cl^- confirmed (01 mark)

(ii) $NH_4Cl_{(s)} \xrightarrow{Heat} NH_{3(g)} + HCl_{(g)}$ (02 marks)

(iii) The Cation was NH_4^+ and anion was Cl^-

(iv) Sample M was NH_4Cl (01 mark)